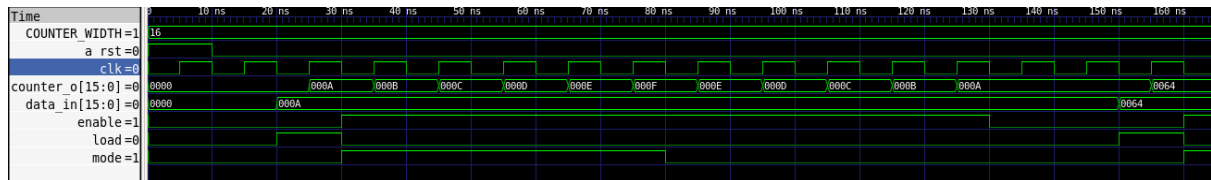


LAB05:

Alunos: Gustavo Santiago Sousa

Resumo sobre o Guia da Synopsys

Simulação:



```
module counter
# (
    parameter COUNTER_WIDTH = 16
)
(
    input  wire clk,
    input  wire a_rst,
    input  wire enable,
    input  wire load,
    input  wire mode,
    input  wire [COUNTER_WIDTH-1:0] data_in,
    output reg [COUNTER_WIDTH-1:0] counter_o
);

always @(posedge clk or posedge a_rst) begin
    if (a_rst) begin
        counter_o <= {COUNTER_WIDTH{1'b0}};
    end else begin
        if (load) begin
            counter_o <= data_in;
        end else if (enable) begin
            counter_o <= mode ? counter_o + 1 : counter_o - 1;
        end
    end
end

endmodule
```

```
`timescale 1ns/1ps
```

```
module counter_tb;
```

```

parameter COUNTER_WIDTH = 16;

reg clk;
reg a_rst;
reg enable;
reg load;
reg mode;
reg [COUNTER_WIDTH-1:0] data_in;
wire [COUNTER_WIDTH-1:0] counter_o;

counter #(
    .COUNTER_WIDTH(COUNTER_WIDTH)
) uut (
    .clk(clk),
    .a_rst(a_rst),
    .enable(enable),
    .load(load),
    .mode(mode),
    .data_in(data_in),
    .counter_o(counter_o)
);

always #5 clk = ~clk;

initial begin
    $dumpfile("counter.vcd");
    $dumpvars(0, counter_tb);

    // init
    clk = 0;
    a_rst = 1;
    enable = 0;
    load = 0;
    mode = 0;
    data_in = 0;

    // reset
    #10;
    a_rst = 0;

    // load valor inicial
    #10;

```

```
load = 1;
data_in = 16'd10;
#10;
load = 0;

// contagem crescente
enable = 1;
mode = 1;
#50;

// contagem decrescente
mode = 0;
#50;

// desabilita contador
enable = 0;
#20;

// novo load
load = 1;
data_in = 16'd100;
#10;
load = 0;

// conta de novo
enable = 1;
mode = 1;
#50;

$finish;
end

endmodule
```